

Name	Description (required)	Priority (1 - highest, 5 - lowest)	Rationale	Author
Account				
1 User Login	The user shall login to the system with the following credentials:		1 Students must be able to log into their unique accounts	Eden M
1.1	- username		1 Username will be unique to the user	Eden M
1.2	- password		1 password created by the user will provide security to the account	Eden M
2 Account Creation	The user shall create an account with the following credentials: (* required)		1 to be able to ensure the student is a USC student and that every student can have their own account	Eden M
2.1	- username (USC username) *		1 unique username will be created for each user	Eden M
2.2	- password (USC password) *		1 password created by the user will keep their data safe	Eden M
2.3	- date of birth *		1 to confirm that they aren't too young to use this software	Eden M
2.4	- email (USC email) *		1 users may forget their password or username, if so they need an email to collect and/or change the password	Eden M
2.5	- USC id *		1 users must be a USC student	Eden M
3 Account details	User accounts shall contain the following information (see each account):		1 Each account needs their own information to effectively progress students, create questions, and manage the system	Marshall P
4 Password Reset	The user shall be able to reset their password		2 Students may forget their password and still want to access their account	Eden M
4.1	User shall be able to reset their password by clicking a "forgot password" button		2 Users, when they forget their password, will need a clear way to access their account.	Eden M
4.2	Clicking "Forgot password" shall send the user an email with a link to reset their password		2 Users who forgot their password need a way to change their password without sacrificing security.	Eden M
5 CAS Authentication	The system shall support user authentication via CAS:		While there will be more than USC users, improving ease of access to account features locally could improve the userbase early on	Marshall P
5.1	- There shall be a "Log in with USC account button" which students shall click to authenticate		2 Use USC's CAS login page to authenticate users and log them into the system	Marshall P
6 Edit Accounts	The user shall edit their own accounts		2 Aspects of a user may change (like the email or username), so the account will need to reflect that.	Marshall P
6.1	User shall go to settings to be able to edit their account		Users need a clear, standard way to change the information in their account, and a settings menu is a standard way of changing account data.	Eden M
7 Settings	There shall be a settings section within the account for the user to manage their data:		2 Users need a special section for settings to logically separate functionality	Eden M
7.1	- add new emails		2 Users may acquire a new email account they want linked to the account.	Eden M
7.2	- modify email		2 Users may acquire a new email account they want linked to the account.	Marcus M
7.3	-resetting password		2 Users need to be able to change their password to improve security.	Eden M
Student				
1 User Log in/out buttons	The user shall log out by clicking a "Log out" button in a corner		1 Students need an easy way to log out of the system	Eden M
1.1	The user shall log in by clicking a "Log in" button in a corner		1 Students need an easy way to log into the system	Eden M
2 User Profile	The user's profile shall be viewable by other users with the following information		3 Students can share their information with others students or potential employers	Ainsley W
2.1	- username		3 Students need to be able to see names to recognize other users by	Ainsley W
2.2	- major		3 Students benefit from interacting with students of the same major	Ainsley W
2.3	- current/past courses taken		3 Students benefit from interacting with students who have taken the same courses	Ainsley W
2.4	- number of questions answered		3 Students benefit from the competition created by showing off the amount of questions they've answered	Ainsley W
3 Posting Answers	The user shall be able to post answers to questions...see Solutions section		1 Students need to be able to respond to questions (ie the main point of the software)	Ainsley W
4 Saving to Student Account	The system shall save a student account's:		1 Student accounts have unique memory needs to be most effective	Ainsley W
4.1	- courses taken		1 This allows students to take on questions that fit the courses they've already taken	Ainsley W
4.2	- completed questions		1 This allows students to track progress and tackle problems that they haven't seen before	Ainsley W
5 Rating a Question	The user shall be able to rate the quality/relevance of a question to its specific subject		3 Based on its rating, a question has the potential of being removed from the system	Ainsley W
5.1	The user shall rate the quality of a question by ranking it from 1 to 5		3 User feedback would enable improvement of the system for future users	Eden M
6 Filter Questions	The user shall be able to filter existing questions...see Filters section		2 This enables the student to drill specific topics, increasing skill	Ainsley W
Editor				
1 Editor Log in/out	The editor user shall be able to login to the software...see Student section		1 This is necessary for the editor to be able to use the system	Marcus S
1.1	The editor user shall be able to logout of the software...see Student section		This is necessary for security, preventing an unattended, shared computer from providing unauthorized access to the system	Marcus S
2 Adding Questions	The editor user shall add a question...see Question section		This creates a crowd sourced question bank, which gives students access to questions that are deemed important by teachers and employers	Ainsley W
3 Deleting Questions	The editor user shall delete a question		1 This keeps the question bank relevant and well-moderated.	Ainsley W
3.1	- must give reason as to why the question was deleted		2 This provides feedback and justification to prevent unnecessary removal	Ainsley W
4 Save to Editor Account	The system shall save an editor account's:		1 Editor accounts have unique memory needs to be most effective	Marshall P
4.1	- questions made		1 This allows access to questions after they are made, permitting deletion and editing.	Marshall P
5 Filter Questions	The editor shall be able to filter existing questions...see Filters section		2 Editors need to have an understanding of what types of questions exist to be able to create pertinent questions.	Ainsley W
6 Rating a Question	The editor shall be able to rate the quality/relevance of a question to its specific subject...see Student section		3 Editor feedback is important to make the system most effective since they are the judges of a student's ability	Eden M
7 View Editor's Account	The editor user's account shall be viewable by other users		3 Editors benefit from seeing student account progress and what other editors are doing.	Ainsley W
7.1	- includes a list of previously posted questions by that editor		Students and editors would benefit from seeing similar problems, likely posted by the same editor. Additionally, editors can be a particular company, allowing students to search for problems from that company.	Ainsley W
Admin				
1 Admin Log in/out	The admin user shall be able to login to the software...see Student section		1 Admins must be able to access the software.	Ainsley W
1.1	The admin user shall be able to logout of the software...see Student section		1 Admins must be able to logout to prevent unauthorized access on a shared device.	Ainsley W
2 Adding Questions	The admin user shall add questions...see Question section		1 Admins should be able to create questions to be able to adequately perform their administrative duties over the questions posted to the system.	Ainsley W
3 Modify User Account	The admin shall modify user accounts by:		1 Admins should be able to use their access to help users with broken or inaccessible accounts as well as moderating the system by adding or removing accounts.	Marcus M
3.1	- deleting student accounts		1 If a student account is acting inappropriately, the admins must be able to remove that problem.	Marcus M
3.1.1	Admin shall click the 'delete' button on a student account		1 There needs to be a way that allows the admin to perform the delete function.	Eden M
3.2 Editing Student Account	- editing student account details		1 Students may add inappropriate information on their accounts. The admins must be able to make that information appropriate.	Marshall P
3.2.1	Admin shall click the 'edit' button on a student account		1 Admins need a way that allows them to perform the edit function on a student account	Eden M
4	The admin shall be able to search student accounts by username		1 Admins need a way to find a student account to edit or delete	Eden M
Question				
1 Making a Question	The editor shall make a question containing the following information:		1 New questions will be relevant over time, so it is important to be able to add more questions	Marcus S
1.1	- title *		1 A quick description for the user	Eden M
1.2	-question text/description*		1 This will be the actual question the users will answer	Marcus S
1.3	- a list of 1 or more tags*		1 So students will be able to filter by these tags to find relevant questions	Eden M
1.4	- a list of 1 or more hints*		2 If students get stuck, they might be able to still solve the problem without seeing the answer	Eden M
1.5	- a difficulty		Students may be new to coding and only want easier questions, or may be more advanced and want to avoid what they know.	Marcus S
1.6	- interview mode		2 Students may want to simulate an interviewing environment as practice for the real thing	Eden M
1.7	- Solution ...see Solution		1 Students will need to know if they got the solution to the problem correct	Marcus S
2 Behavioral Question Type	The user shall be presented with multiple choice questions for behavioral questions		2 Students will need to be able to practice behavioral interview questions	Marcus S
3 Conceptual Question Type	The user shall be presented with multiple choice questions for conceptual questions		1 Students will need to practice for conceptual questions that may be asked in interviews	Marcus S
4 Technical Question Type	The user shall be presented with types of technical problems:		1 Students will need to practice for technical problems that they may be presented with in interviews	Marshall P
4.1	- Predict the output		1 Some technical interview problems will ask a student to predict the output of a particular snippet of code	Marshall P
4.2	- Fill in the blank		1 Some technical interview problems will ask a student to fill in the blank in a statement about code	Marshall P
4.3	- Multiple choice on behavior		1 Some technical interview problems will ask a student to choose a particular behavior out of a lineup	Marshall P
4.4	- Multiple choice on best approach		1 Some technical interview problems will ask a student to choose a particular approach out of a lineup	Marshall P
4.5	- Error spotting		1 Some technical interview problems will ask a student to spot an error in a particular code snippet.	Marshall P
Filters				
1 Filter by Tag	The user shall filter the list of questions by the following tags ...see Question :		2 Students should be able to find questions relevant to what they need to practice, and filter out those that they don't want	Eden M
1.1	- Course Number		2 Students that have completed different courses have different levels of knowledge in coding	Eden M
1.2	- Difficulty		2 Students may want more difficult or hard questions depending on their skill level	Eden M
1.3	- discipline:		2 Students may want different types of questions depending on their area of expertise	Eden M/Marshall P
1.3.1	-computer science		2 Students may need to drill computer science problems	Eden M/Marshall P
1.3.2	-computer information systems		2 Students may need to drill computer information systems problems	Eden M/Marshall P
1.3.3	-computer engineering		2 Students may need to drill computer engineering problems	Eden M/Marshall P
1.3.4	-cyber security		2 Students may need to drill cyber security problems	Eden M/Marshall P
1.3.5	-data science		2 Students may need to drill data science problems	Eden M/Marshall P
1.4	- Question Type (technical, behavioral, conceptual)		2 Students may be focusing on different aspects of the interview	Marcus S
1.5	- Approval Rating		3 Students may want to work on questions promoted by other students due to popularity or relevance	Marcus S
2 Filter Drop Down	The user shall filter the list of questions by using a drop down menu		2 Students will need a way to access the filter	Eden M
Solution				
1 Solution Interaction	The user shall interact with a question's solution by:		1 Students will need to interact with a question to be able to answer and get the most out of this application	Marcus S
1.1	The user shall view the list of solutions for a technical question.		1 Students will need to know the solution to a question to improve and know if their solution was correct.	Marshall P
1.1.1	- pseudocode solutions		2 So that students can understand the concepts that are larger than the language they are coding in	Marshall P
1.1.2	- Different approaches		2 Adding different approaches gives a student a broader understanding of how to solve problems	Marshall P
1.2	The user shall either approve or disapprove of a solution		3 Students will be able to see which answer is the most correct by seeing how much approval it has	Marcus S
1.2.1	User shall either approve or disapprove a solution by hitting either a thumbs up or a thumbs down		3 This is an intuitive way for students to show their approval or disapproval of an answer	Eden M
1.4	- User shall be able to access hints by clicking the "hint" button		3 Beginner students may not be able to solve a problem and they would need a bit of a push to find the solution.	Eden M
1.4.1	-Clicking this hint button shall show the hint provided by the creator...See Question section		3 Creators know best how to create hints, and students will need hints.	Eden M
2 Posting Solutions	Students should post their solution by:		1 Students will need to have a way to post their solutions to the system	Marshall P
2.1	- selecting whether or not to add their profile to their solution		3 Students will need a way to share their profiles with their solutions	Marshall P
2.2	- posting a text file		1 Students need to be able to post their solutions in a way that is easy to implement.	Marshall P
* = required for user to fill out to complete action				